

Multiple Choice (10 marks)

1. If $f(z)$ is an analytic function that maps the entire finite complex plane into the real axis, then the imaginary axis must be mapped onto
 - (a) a circle
 - (b) an infinite interval
 - (c) the negative real axis
 - (d) the positive real axis
 - (e) a ray
2. The original price of a new bicycle is \$138.00. If the bicycle is marked down 15%, what is the new price?
 - (a) \$125.00
 - (b) \$20.70
 - (c) \$130.00
 - (d) \$133.00
 - (e) \$115.00
3. Consider the initial value problem $y' + \frac{1}{4}y = 3 + 2 \cos 2t$, $y(0) = 0$. Determine the value of t for which $y(t) = 11$.
 - (a) 11.234567
 - (b) 10.065778
 - (c) 6.789012
 - (d) 7.654321
 - (e) 9.876543
4. Write the ratio of 96 runners to 216 swimmers in simplest form.
 - (a) 16 over 36
 - (b) 9 over 4
 - (c) 8 over 18
 - (d) 48 over 108
 - (e) 24 over 54
5. In the group $G = \{2, 4, 6, 8\}$ under multiplication modulo 10, the identity element is
 - (a) 9
 - (b) 7

- (c) 8
- (d) 1
- (e) 4

True / False (10 marks)

Answer True or False

6. True or False: To ensure at least two people share a birth month, a minimum of 24 individuals must be included in the group.
7. True or False: The cumulative distribution function of the standard Gaussian distribution is not log-concave and is instead linear.
8. True or False: The double integral of the function $f(x,y)=x*y^3 \exp\{x^2+y^2\}$ over $I=[0,1] \times [0,1]$ evaluates to approximately 0.4295.
9. True or False: A customer can order a pizza with 3 toppings from 8 available options in 36 different ways.
10. True or False: If a number is 12, then eighty-five more than its square equals the square of the quantity that is 17 less than the number.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. For a certain natural number n , n^2 gives a remainder of 4 when divided by 5, and n^3 gives a remainder of 2 when divided by 5. What remainder does n give when divided by 5?
12. Given the equation $|x - 4| - 10 = 2$, analyze the solutions for x and discuss the implications of these solutions by calculating the product of all possible values of x .

End of Paper